Experimental setup

A. Simulation harness — USED FOR ALL COMMITTED RESULTS

Unmodified firmware logic

smart_home.ino compiled against host stubs for Arduino, FreeRTOS, WiFi, MQTT, NVS and the OLED. The firmware source is not altered for testing.

Campaign

6,000 trials
seed base 1000
rates 0 to 20,000 msg/s
fork() per trial

Fuzzer

3,000,000 iters
seed 7, 6 classes
ASan + UBSan
oracle: no state change, no counter

Limitation that defines the open step

The harness advances a logical clock. There is no radio, no true parallel execution and no CPU contention, so latency variance is zero.

Analysis

analyse.py: aggregation, Mann-Whitney U with tie correction, standard library only. make_figures.py: validated palette.

B. Hardware rig — SPECIFIED, NOT YET RUN

Device under test

ESP32 DevKit
running v2.2.0
unmodified

GPIO

Measuring board

second ESP32,
independent clock,
logic probe

Four conditions

A.	idle (control)	no RF isolation
B.	legitimate network load	no RF isolation
C.	command-interface stress	no RF isolation
D.	radio-layer stress	AUTHORISED ISOLATED RF ONLY

Dependent variable

Physical sensor-to-siren latency distribution, compared across conditions with the same non-parametric test (analyse_hw.py).

A to C need two boards and a probe. No attack tooling is published.